

CATHKIN ESTATES INFRASTRUCTURE REVIEW

Road Upgrade Kick Off

21st July 2026

Objectives & Agenda

Road Infrastructure Repair & Sub-Base Reinstatement Strategy



Objectives

- 1. Assessment of Options:** Technical evaluation across flexible seals, Cape Seals, pavers, and reinforced concrete options.
- 2. Action Plan:** Formulate a structured, phased operational rollout utilizing specialized external contractor support and civil engineering oversight[cite: 1].



Agenda

- **Foundation & Budget:** Sub-base preparation rules and R1M budget coverage.
- **Issues & Solutions Review (Round-Table):** Open discussion on current road failures (e.g., concrete cracking, clay sub-base movement) and proposed options for consideration.
- **Comparative Feasibility:** Single-access logistics, weather risks, contractor fit, and lifecycle costs.
- **Action Plan & BOQs:** Technical appendices, financial breakdowns, and rollout phases.

Flexible vs. Rigid Pavement Pathways

Comparing how flexible sprayed seals versus rigid paver/concrete slabs distribute heavy vehicle loads



Flexible Seals (Asphalt, Chip & Spray, Cape Seal)

Possess zero bridging strength. If the underlying sub-base shifts even 5mm under a 10-cube cement truck, the surface splits instantly. Mandatory sub-base stabilization is required to avoid severe crack propagation.



Rigid Systems (Pavers & Cast Concrete)

Transfer weight across high structural beam and joint shear capacity. Reinforced cast concrete and interlocking pavers bridge soft clay spots effectively, reducing deep subgrade excavation requirements in steep zones.

Overview of Surfacing Options

Defining the five primary technical methodologies evaluated for Cathkin Estates

Double Chip & Spray

Bituminous binder sprayed onto base, followed by two sequential stone chip layers rolled to lock aggregate mosaic.

Cape Seal (Colas Spec)

Single-chip aggregate layer (13.2/19mm) sealed and locked with a rich bitumen slurry overlay to resist tire shear.

Hot-Mix Asphalt

Continuous 30mm factory hot asphalt mat laid via paver screed and steel rolled for a smooth riding surface.

60mm Interlocking Pavers

High-strength (35 MPa) precast blocks laid in 90° Herringbone pattern over river sand bed with side concrete kerbs.

In-Situ Reinforced Concrete

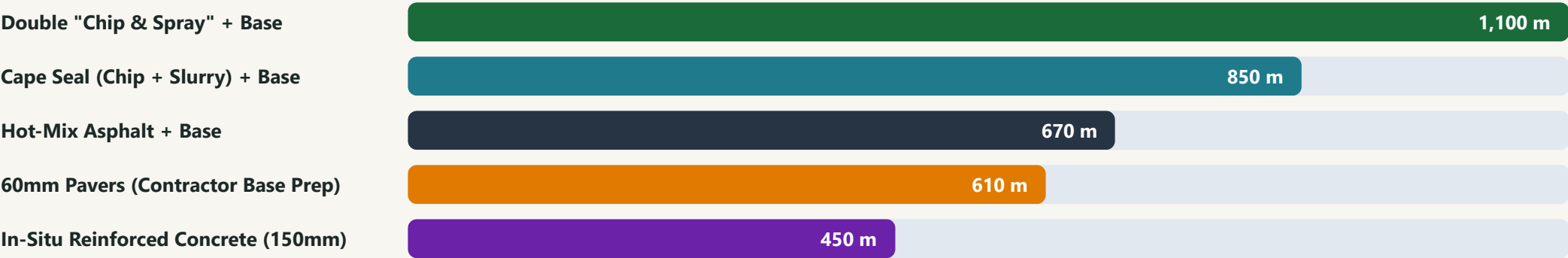
Heavy-duty 150mm slab (30 MPa) cast with Ref 193 steel mesh reinforcement and dowelled saw-cut joints for steep inclines.

Option for Consideration

Combine **Pavers** for general flat corridors with **Concrete** on steep slopes and water crossings—subject to traffic bypass viability, proper sub-base prep, and engineered expansion joints to mitigate cracking.

Sub-Base Realities & Budget Reach

R1M Budget Coverage Comparison Across All Evaluated Options



Note: While cast concrete has a lower initial distance reach (~450m/R1M), its 35+ year lifespan and complete immunity to water ingress make it the ultimate long-term investment for critical steep gradients.

Lifecycle Cost Comparison Matrix

Comparative initial outlay vs. 10-Year, 20-Year, and 30+ Year Total Cost of Ownership (per m²)

Road Option	Surface / m²	Base Prep / m²	Initial Outlay	Lifespan	10-Year TCO / m²	20-Year TCO / m²	30+ Year TCO / m²
Double Chip & Spray	R90 – R140	R150 – R250	R240 – R390	7 – 10 Yrs	R380 – R520 ¹	R640 – R900 ²	R1,000+ (Rebuild) ³
Cape Seal (Colas Spec)	R180 – R290	R150 – R200	R330 – R490	10 – 12 Yrs	R380 – R540 (Lowest 10Y)⁴	R580 – R840 ⁵	R900 – R1,200 ³
Hot-Mix Asphalt	R250 – R350	R150 – R250	R400 – R600	15 – 20 Yrs	R450 – R680 ⁶	R620 – R880 ⁷	R1,100+ (Milling) ³
60mm Pavers (35 MPa)	R300 – R400	R150 – R250	R450 – R650	20 – 30 Yrs	R480 – R680 ⁸	R500 – R750 (Lowest 20Y)⁹	R550 – R800 ¹⁰
Concrete (150mm / 30 MPa)	R400 – R550	R150 – R250	R550 – R800	35 – 50+ Yrs	R560 – R820 ¹¹	R570 – R850 ¹¹	R570 – R850 (Lowest 30Y+)¹²

*** TCO Maintenance Assumptions & Intervention Schedules:**

¹ **Double Chip (10Y):** Includes 1x mandatory fog spray / re-seal cycle at Year 6 + routine crack patching.

³ **30+ Year Rebuild:** Flexible bituminous seals reach end-of-life and require complete milling & structural base rebuilding.

⁵ **Cape Seal (20Y):** Includes 1x full slurry overlay / re-seal cycle at Year 12 + routine edge maintenance.

⁷ **Asphalt (20Y):** Includes 1x thin asphalt overlay or micro-surfacing treatment at Year 15.

⁹ **Pavers (20Y Leader):** Lowest TCO over 20 years on standard flat/general estate road sections.

¹¹ **Concrete (10Y/20Y):** Maintenance limited strictly to elastomeric joint sealant replacement every 8–10 years.

² **Double Chip (20Y):** Includes 2x full re-seals (Y6, Y14) + localized base repairs under heavy truck traffic.

⁴ **Cape Seal (10Y):** Includes minor crack sealing and 1x slurry dilution rejuvenation spray at Year 8.

⁶ **Asphalt (10Y):** Includes routine asphalt joint sealing and localized pothole/edge repairs.

⁸ **Pavers (10Y/20Y):** Includes periodic joint sand sweeping/replenishment & localized edge beam fixes. Zero re-sealing required.

¹⁰ **Pavers (30Y+):** May require minor lifting & re-bedding in soft clay spots, but blocks are fully reusable.

¹² **Concrete (30Y+ Leader):** Undisputed lowest TCO over 30–50 years. Zero structural resurfacing required.

*** Base Prep Variance:** Cape Seal base prep (R150–R200/m²) utilizes targeted BSM (Bitumen Stabilised Material) blending on existing sub-base rather than full-depth aggregate replacement.

Comprehensive Feasibility Matrix

Multi-Criteria Engineering & Logistical Evaluation across all 5 Options

Metric	Double Chip & Spray	Cape Seal	Hot-Mix Asphalt	60mm Pavers	Reinforced Concrete
Single-Access Logistics ¹	Moderate: Short road closures needed.	Moderate: Slurry curing breaks.	High: Cold joints if trucks delay.	Zero Risk: Immediate drive-on.	Managed: Half-width 24h rapid-set.
Weather Dependency ²	High: Rain ruins adhesion.	Moderate: Slurry needs dry day.	Extreme: Wind kills compaction.	None: Pauses during rain.	Low: Rain protection during cure.
Contractor Compatibility ³	Requires specialist bitumen distributor rigs.	Standard civil road contractor scope.	Requires commercial paving equipment.	High: Turnkey paving subcontractor.	High: Turnkey concrete contractor.
Traffic Resiliency ⁴	Infeasible: Tippers shear stone.	Good: Slurry binds stone layer.	Moderate: Summer tire rutting.	Exceptional: Handles turning trucks.	Maximum: Structural slab strength.
Project Verdict ⁵	INFEASIBLE	VIABLE HYBRID	HIGH RISK	OPTIMAL (FLAT/GENERAL)	OPTIMAL (STEEP/BRIDGES)

* Engineering & Operational Commentary:

¹ **Single-Access Logistics:** Single-road access requires continuous resident availability. Pavers allow zero-delay traffic. Concrete uses rapid-hardening cement (CEM I 52.5R) in half-width pours to reopen lanes within 24–48 hours. Asphalt risks cold joints if transit trucks stall.

³ **Contractor Compatibility:** Evaluates external contractor capacity and equipment requirements. Contractor team utilizes heavy graders, rollers, and specialist machinery for 100% of sub-base and surface preparation work.

⁵ **Verdict Rationale:** A hybrid strategy is recommended: **60mm Pavers** for main flat corridors (zero access disruption) and **150mm Reinforced Concrete** for steep inclines, stormwater channels, and bridge approaches[cite: 1].

² **Weather Dependency:** Drakensberg rain and cold snaps severely threaten sprayed bitumen emulsions (chip seals and asphalt cooling). Pavers and cast concrete can be halted or protected with plastic sheeting during rain without ruining laid sections.

⁴ **Traffic Resiliency:** Heavy 10-cube cement trucks (8.2t axle load) exert severe shear force on steep inclines and sharp turns. Rigid surfaces (pavers and concrete) resist subgrade rutting and edge cracking far better than flexible bituminous seals.

SUPPLEMENTARY DOCUMENTATION

Technical Appendix

Detailed bills of quantities, material specifications, and engineering breakdowns supporting the estate road infrastructure evaluation. Based on available reference data and intended for scoping purposes only.

Executing Double Chip & Spray

Sequential On-Site Implementation Workflow

1

Base Prep (Contractor)

Rip soft spots using heavy plant equipment. Stabilize gravel with 42.5N cement, water, shape crown, and compact with roller[cite: 1].

2

Prime Coat

Broom dust off base[cite: 1]. Spray fluid bitumen primer to soak 10mm into base, creating a solid waterproof seal[cite: 1].

3

Bitumen & Stones

Spray hot polymer bitumen[cite: 1]. Drop 13.2mm stone chips and roll[cite: 1]. Repeat with secondary binder and 6.7mm stones[cite: 1].

4

Keying & Brooming

PTR pneumatic rollers knead stones into mosaic[cite: 1]. Restrict traffic for 3 days, then sweep away loose aggregate[cite: 1].

Key Operational Notes

- **Equipment Required:** Heavy Grader/Tractor, Bitumen Distributor Truck, Chip Spreader, PTR Rubber Roller[cite: 1].
- **Traffic Control:** 24h lane restrictions post-application[cite: 1]; 3-day restriction before heavy trucks.
- **Quality Gate:** Base compaction must reach 98% Mod AASHTO before priming[cite: 1].
- **Weather Window:** Requires dry weather and ambient surface temp > 20°C.

Executing Cape Seal (Colas Spec)

Sequential On-Site Implementation Workflow

1

Base Prep (Contractor)

Stabilize existing base using civil earthmoving equipment with cement blending to fix deep-lift failures and shape drainage crowns[cite: 1].

2

Single Chip Layer

Spray heavy polymer binder emulsion and drop single 13.2mm or 19mm aggregate stone chips[cite: 1]. Roll with rubber roller[cite: 1].

3

Slurry Seal Overlay

Squeegee bitumen slurry into chip voids[cite: 1]. Encapsulates stones and seals surface against water ingress[cite: 1].

4

Cure & Smooth Finish

Allow slurry to break from brown to jet black[cite: 1]. Final rolling creates a smooth riding surface resistant to scuffing[cite: 1].

Key Operational Notes

- **Equipment Required:** Earthmoving Plant, Pneumatic Roller, Specialist Squeegee Crew / Slurry Mixer Rig[cite: 1].
- **Cure Time:** 4–6 hours for slurry break before opening to light vehicle access[cite: 1].
- **Quality Gate:** Ensure base moisture < 50% OMC prior to spraying binder[cite: 1].
- **Weather Window:** Requires ambient temp > 15°C and zero rain forecast within 12h[cite: 1].

Executing Hot-Mix Asphalt

Sequential On-Site Implementation Workflow

1

Base Prep (Contractor)

Mandatory sub-base stabilization using contractor machinery to rip failing spots, blend cement, wet, shape, and compact[cite: 1].

2

Tack Coat Glue

Sweep cement base clean[cite: 1]. Spray light tack coat of liquid bitumen emulsion to act as chemical glue[cite: 1].

3

Paving & Roll

Discharge ~140°C asphalt into paver hopper[cite: 1]. Screed continuous 30mm mat and roll immediately behind screed[cite: 1].

4

Kneading & Cooldown

Run heavy rubber pneumatic rollers to compact aggregates and seal surface pores before opening to traffic[cite: 1].

Key Operational Notes

- **Equipment Required:** Heavy Earthmoving Plant, Mechanical Asphalt Paver, Steel Drum Roller, PTR Roller[cite: 1].
- **Thermal Window:** Asphalt must arrive > 130°C from Estcourt plant[cite: 1]; cold joints form if delays occur[cite: 1].
- **Cure Time:** Surface cools in 2–4 hours for immediate vehicle access.
- **Quality Gate:** Strict compaction density checks required to prevent premature rutting[cite: 1].

Executing Interlocking Pavers

Sequential On-Site Implementation Workflow

1

Base Prep (Contractor)

Rip failed clay spots using heavy plant equipment. Backfill with stable soil, add cement to soft zones, and heavily compact[cite: 1].

2

Edge Kerbs & Sand

Cast thick concrete edge beams on both sides[cite: 1]. Screed a precise 20mm–25mm dry, washed river sand bedding layer[cite: 1].

3

Herringbone Laying

Lay heavy 60mm (35 MPa) interlocking bricks in a tight 90° Herringbone pattern to resist braking shear forces[cite: 1].

4

Locking & Vibration

Sweep dry silica sand into joint gaps[cite: 1]. Run plate compactor to vibrate sand deep into joints, locking blocks[cite: 1].

Key Operational Notes

- **Equipment Required:** Heavy Earthmoving Plant, Screed Rails, Heavy Plate Compactor with rubber mat[cite: 1].
- **Zero Access Risk:** Cars can drive over laid sections immediately[cite: 1]; ideal for single-road zones[cite: 1].
- **Turnkey Execution:** High match for specialist paving sub-contractor Teams[cite: 1].
- **Quality Gate:** Lateral concrete kerbs must cure 48h prior to block vibration[cite: 1].

Executing In-Situ Reinforced Cast Concrete

Sequential On-Site Implementation Workflow

1

Half-Width Excavation

Box out half roadway to 150mm depth using contractor earthmoving equipment[cite: 1]. Fix side shutters to establish exact fall and cross-fall gradients[cite: 1].

2

Subgrade & Mesh Ref 193

Compact subgrade, lay 250-micron plastic DPM, and set steel Mesh Ref 193 with dowel bars for lateral joint transfer[cite: 1].

3

Rapid-Set 30 MPa Pour

Pour CEM I 52.5R 30 MPa ready-mix concrete, vibrate, beam screed, and finish with broom texture for wet tire grip[cite: 1].

4

Joints & 24h Opening

Saw-cut expansion joints within 12h, seal with elastomeric silicone, and open lane to light cars within 24–48 hours[cite: 1].

Key Operational Notes

- **Equipment Required:** Civil Earthmoving Plant, Timber/Steel Shutters, Concrete Poker Vibrator, Beam Screed[cite: 1].
- **Single-Road Management:** Half-width pours allow resident bypass traffic on open lane[cite: 1].
- **Target Zones:** Optimal for steep inclines, drainage channels, and bridge approaches[cite: 1].
- **Quality Gate:** Saw-cut expansion joints must be cut within 12 hours to prevent random cracking[cite: 1].

Technical Appendix: Double Chip & Spray BOQ

Item Category	Work Description / Materials Spec	Sourcing / Resource	Low Cost / m ²	High Cost / m ²
Base Prep (Stage 1)	Contractor clay excavation and profiling to 2% cross-fall crown	External Civil Contractor	R40.00	R70.00
Base Prep (Stage 2)	Supply & blending of Grade 42.5N cement for stabilization	Materials Procurement	R60.00	R100.00
Base Prep (Stage 3)	Moisture stabilization and vibratory soil compaction	Civil Contractor & roller	R30.00	R55.00
Prime Application	Surface dust clearing and spraying fluid bitumen primer	Specialist Subcontractor	R20.00	R35.00
1st Aggregate Seal	Hot polymer binder spray & 13.2mm stone chip laying	Specialist Subcontractor	R40.00	R65.00
2nd Aggregate Seal	Secondary hot binder spray & 6.7mm gap-locking stone chips	Specialist Subcontractor	R30.00	R45.00
Finish & Rolling	Pneumatic rubber roller compacting & sweeping	Subcontractor / Crew	R20.00	R20.00
TOTAL OUTLAY	Complete Double Chip & Spray road system	Contractor Execution	R240.00 / m ²	R390.00 / m ²

Technical Appendix: Cape Seal BOQ (Colas Spec)

Item Category	Work Description / Materials Spec	Sourcing / Resource	Low Cost / m ²	High Cost / m ²
Base Stabilization (BSM)	Deep-lift patching, cement/foamed bitumen blending, grading	Civil Contractor & Plant	R100.00	R140.00
Foundation Compaction	Heavy rolling, moisture conditioning, and priming	Contractor Plant & roller	R50.00	R60.00
Single Chip Seal Layer	Polymer emulsion spray and 13.2mm/19mm chip spread	Turnkey Subcontractor (Colas)	R90.00	R140.00
Bitumen Slurry Overlay	Bitumen slurry squeegee application into chip voids	Turnkey Subcontractor (Colas)	R90.00	R150.00
TOTAL OUTLAY	Complete Cape Seal System with Base Stabilization	Contractor Execution	R330.00 / m ²	R490.00 / m ²

Technical Appendix: Hot-Mix Asphalt BOQ

Item Category	Work Description / Materials Spec	Sourcing / Resource	Low Cost / m ²	High Cost / m ²
Sub-base Preparation	Ripping old subgrade, blending, grading, compacting	Civil Contractor Plant	R30.00	R50.00
Stabilizing Base	Cement Treatment Base agent blending (5% cement)	Materials Procurement	R60.00	R90.00
Foundation Compaction	Watering, rolling, and grading target areas	Contractor Plant & roller	R30.00	R50.00
Tack Coat Glue	High-pressure sweeping and liquid tack coat spray	Specialist Subcontractor	R15.00	R30.00
Asphalt Materials	Medium-grade continuous hot asphalt mix (Estcourt)	Contractor Materials	R160.00	R220.00
Transport & Logistics	Insulated thermal bulk transport premium from Estcourt	Contractor Transport	R65.00	R100.00
Mat Laying & Rolling	Mechanical paver machine screeding 30mm thick mat	Specialist Subcontractor	R40.00	R60.00
TOTAL OUTLAY	Premium 30mm Hot-Mix Asphalt overlay surface	Contractor Execution	R400.00 / m ²	R600.00 / m ²

Technical Appendix: 60mm Interlocking Pavers BOQ

Item Category	Work Description / Materials Spec	Sourcing / Resource	Low Cost / m ²	High Cost / m ²
Deep Foundation Fixes	Spot ripping, deep clay excavation, base preparation	Civil Contractor / Earthmoving Plant	R150.00	R250.00
Lateral Edge Beams	Casting flush in-situ concrete kerbs / haunching	Paving Subcontractor	R45.00	R75.00
Screeded Sand Bed	Procurement and screeding washed river sand (25mm)	Materials Supply & Contractor	R35.00	R55.00
Paver Material Supply	60mm structural concrete blocks (35 MPa SABS)	Materials Procurement	R220.00	R270.00
TOTAL OUTLAY	35 MPa Interlocking Pavement system	Contractor Execution	R450.00 / m ²	R650.00 / m ²

Technical Appendix: In-Situ Reinforced Concrete BOQ (150mm Slab)

Item Category	Work Description / Materials Spec	Sourcing / Resource	Low Cost / m ²	High Cost / m ²
Subgrade Prep & Boxing	Contractor excavation to 150mm depth, leveling, compaction	Civil Contractor / Earthmoving Plant	R40.00	R70.00
Formwork & DPM	Side shutter fixing, 250um DPM underlay, & 16mm dowel bars	Civil Concrete Contractor	R45.00	R65.00
Steel Mesh Reinforcement	Ref 193 welded steel fabric mesh & support spacers	Materials Supply	R65.00	R95.00
30 MPa Rapid-Set Concrete	30 MPa concrete (CEM I 52.5R), placing, & vibration	Local Batch Plant / Contractor	R340.00	R450.00
Joints & Finishing	Broom finishing, 3m saw-cut joints, & elastomeric sealer	Contractor / Specialist	R40.00	R60.00
TOTAL OUTLAY	Complete Heavy-Duty 150mm Reinforced 30 MPa Concrete Slab	Contractor Execution	R530.00 / m ²	R740.00 / m ²

Reference Material Summary

Historical Contracts & Audits

JAVSEAL (2017): Establishes baseline pricing & environmental rules.

Gert Rautenbach & Colas (2021–2022): Audits recommended targeted deep-lift patching and Cape Seals to prevent premature base failure.

Strategic & Commercial Specs

Manager Brad Mack & BoQ Sheets: Area calculations, clay tilling, sub-base requirements, and traffic weighting.

Action Asphalt Estimates (2022–2023): Gate repairs & sub-soil drainage crossings at ~30m intervals.